

BHANU SHARMA

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TECHNICAL SKILLS

Programming Languages: Python, R, SQL, UNIX/Shell, Scala, Java, SAS

Libraries: Pytorch, Tensorflow, Numpy, OpenCV, scikit-learn, Pyspark, Pandas, Matplotlib, ggplot2,

AI/ML Tools: vLLM, TRT-LLM, LangChain, LangGraph, Vector Databases, AIPerf, Zeus Energy, AWS Bedrock

Tools: Jupyter, Power BI, Tableau, Azure, Databricks, Snowflake, Spark, Docker, ADF, AWS, MS Office, Github

Skills: Machine Learning, AI, Natural Language Processing (NLP), Transformers, Agentic AI, Benchmarking, AI Workflows, Data Visualization, Big Data, Data Warehousing, Business Intelligence, ETL pipelines, CI/CD, Agile, Cloud

EXPERIENCE

Dell Technologies, Hopkinton, MA: AI Research Intern

Jan 2026 - June 2026

- Developed LLM inference simulator that, given hardware/model specs and SLA targets, evaluates serving configs to predict throughput, latency, TTFT, and ITL within $\pm 15\%$ **accuracy at batch sizes of 512+** giving optimal config for vLLM serving setup, incorporating continuous batching, parallelism (TP, DP, PP), and disaggregated serving.
- Extended the simulator to support **Mixture-of-Experts (MoE)** models, EP, and multiple quantization strategies, enabling efficient and cost-aware inference optimization saving a projected **1500+ GPU hours**
- Modeled and Implemented Novel LLM Inference Energy and Power estimator based on roofline model of energy, achieving up to **90% accuracy**

Northeastern University, Boston, MA: Teaching Assistant

Sept 2025 - Present

- CS 6140: Machine Learning** - Supported **87+ graduate students** with core ML concepts by holding weekly office hours and grading
- MATH 7340: Statistics for Bioinformatics** - Supported **90+ graduate students** with R programming and core statistical concepts with weekly office hours

UnitedHealth Group (Optum), Gurgaon, India: Data Engineer

Aug 2020 - Jun 2024

- Migrated legacy ETL pipeline to Spark/Scala and Snowflake on Azure, reducing database **costs by up to 90%**
- Redesigned ETL architecture using PySpark, MongoDB, Azure Databricks, and ADF, handling **3TB+ of daily data**
- Built Power BI dashboards for **50+ clients** and a Python KPI alerting system saving **6+ hours/week**

PUBLICATIONS

- The Hidden Thirst of AI: Estimating Freshwater Consumption per LLM Query** - [DOI](#) Sept 2026
Green (MDPI), Vol. 1, Issue 2
- Operational Resilience Under Carbon Constraints: A Multi-Agentic Approach** - [DOI](#) March 2026
Systems (MDPI), Vol. 14, Issue 4
- Some Patterns of Sleep Quality and Daylight Saving Time Across Countries** - [DOI](#) Aug 2025
Int. Journal of Data Mining & Knowledge Management Process (IJDKP), Vol. 15, No. 4

PROJECTS

TerrierTA - Multi-Agent AI Teaching Assistant Framework - [Github](#)

July 2026

- Built a multi-agent RAG framework using LangGraph to unify rubric-grounded grading, citation-backed student Q&A, and content generation with human-in-the-loop review gates, for university-level courses

Swaad - Flavor Profile Based Food Recommendation System - [Github](#)

Dec 2025

- Built an AI food recommendation system using Gemini Flash multimodal LLM to extract dishes from menu images, generate per-dish embeddings, and apply KNN matching against a 1,000+ ingredient user taste profile in real time.

EDUCATION

MS in Bioinformatics (Data Science Focus) (GPA: 4.0)

Sept 2024 - Dec 2026

Northeastern University

Boston, MA, USA

Relevant coursework: Deep Learning, Foundations of AI, ML, Statistics, Algorithms

Bachelor of Technology in ECE (GPA: 7.7/10)

Aug 2016 - May 2020

Punjab Engineering College (PEC)

Chandigarh, India